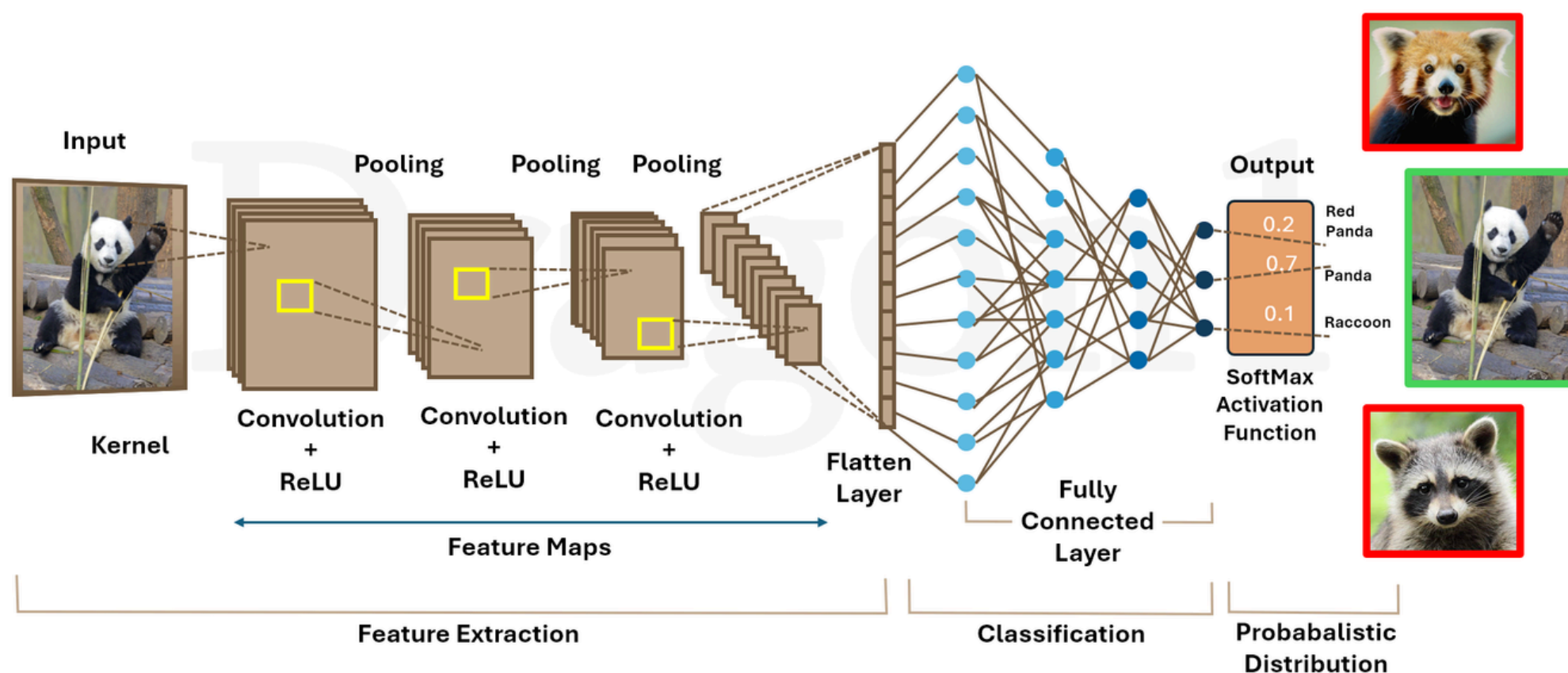


DAY 3 OF DAILY DEEP-DIVE SERIES

Convolutional Neural Network

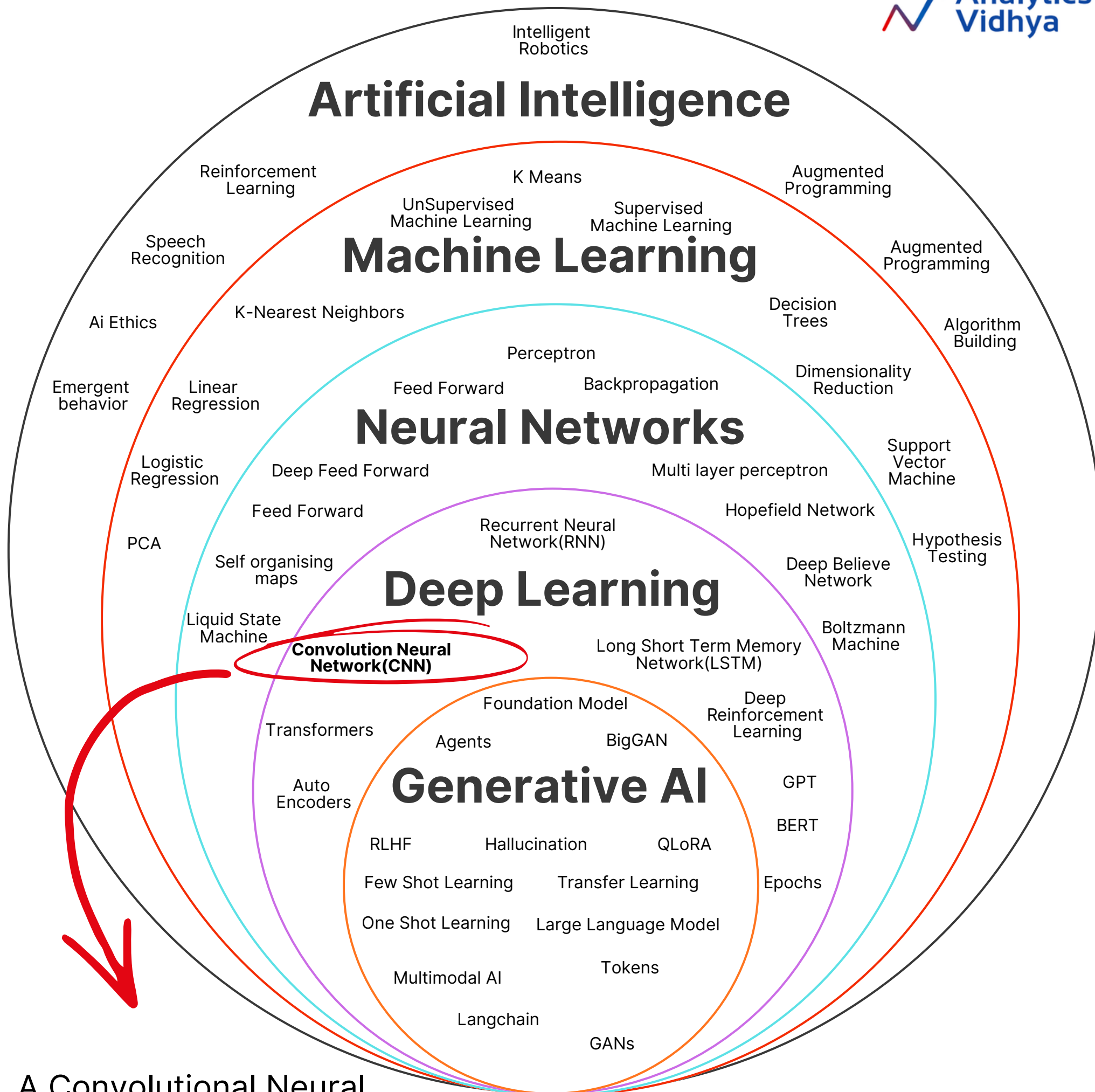
The deep learning model that taught **machines to recognise images**



CNN pipeline:



Your eyes do this in 0.013 seconds.
A CNN does it too. Swipe to see how.



A Convolutional Neural Network is a type of deep learning model designed to process visual data. Images, videos, anything pixel-based.

What exactly is a CNN?

A Convolutional Neural Network is a type of deep learning model designed to process visual data. Images, videos, anything pixel-based.

THINK OF IT THIS WAY

A regular neural network sees an image as a long list of numbers and loses all spatial information.

A CNN looks at small patches at a time, just like your eyes do.

WHERE IS IT USED?

Image recognition

Photos, faces, objects

Medical imaging

X-rays, MRI scans

Self-driving cars

Road, sign detection

Video analysis

Action recognition

Art generation

Style transfer, GANs

Satellite imagery

Maps, land detection

The inspiration

The human eye

CNNs were directly inspired by how the human brain processes visual information.

HOW YOUR EYES WORK

Step 1 — Your eye detects edges first

Not the whole image at once. Small local regions only.

Step 2 — The brain combines edges into shapes

Corners become outlines. Outlines become objects.

Step 3 — The brain matches the pattern to memory

Shape plus fur plus four legs equals a dog.

A CNN does the exact same thing

It scans patches, detects edges, builds shapes, then recognises the full object.

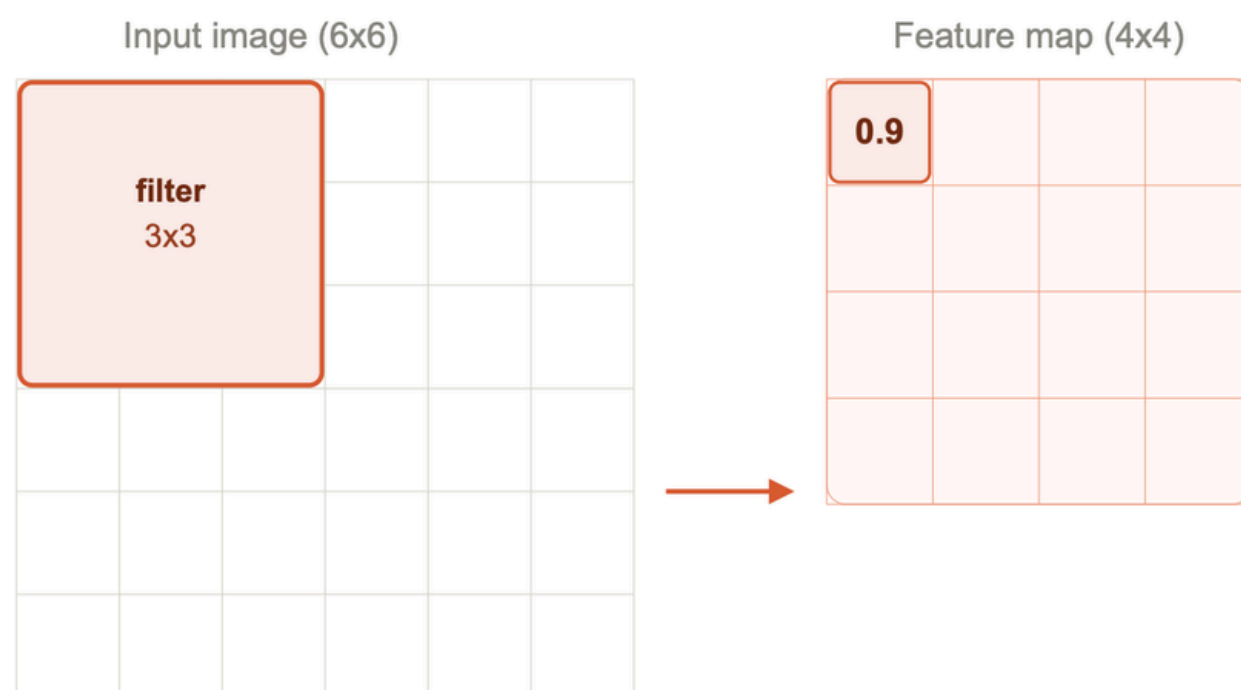
The convolution

How CNN scans an image

A small window called a filter slides across the image, scanning every patch.

At each position it asks: "Is this feature here?"

THE FILTER SLIDING ACROSS AN IMAGE



WHAT DIFFERENT FILTERS DETECT

Edge filter
Detects boundaries

Blur filter
Smooths image noise

Sharpen filter
Highlights fine detail

A CNN learns hundreds of these filters automatically during training.

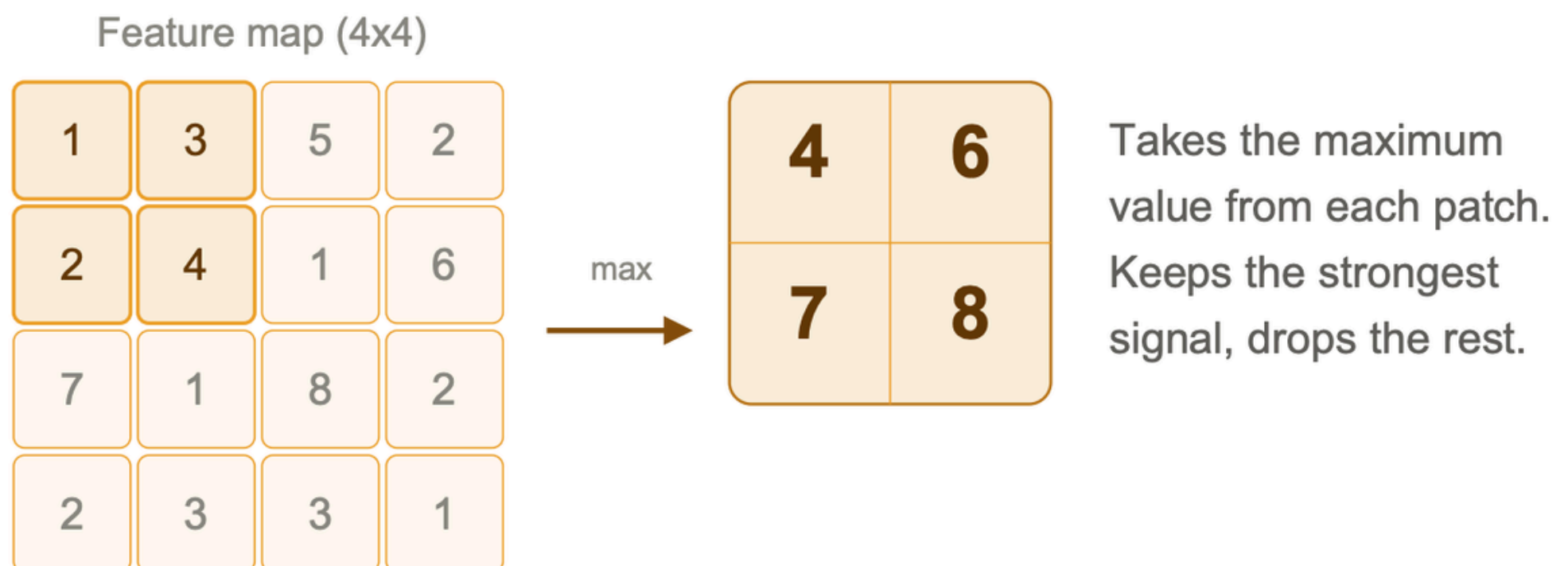
Pooling

Shrink without losing meaning

After convolution, the feature map is still large. Pooling compresses it by keeping only the most important values.

MAX POOLING IN ACTION

After convolution, the feature map is still large. Pooling compresses it by keeping only the most important values.



WHY POOLING MATTERS

Reduces size

Faster training,
less computation

Keeps key info

Dominant features
stay preserved

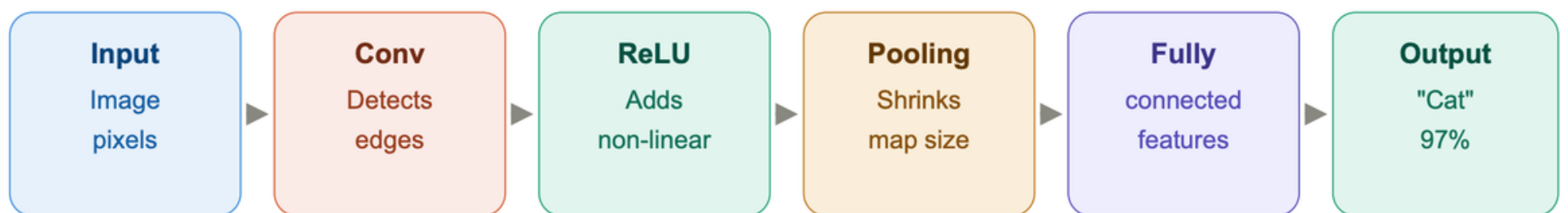
Adds robustness

Small shifts in image
will not fool the model

The full CNN

Architecture

Here is how all layers stack together to turn a raw image into a final prediction.



WHAT EACH LAYER LEARNS

Layer 1

Simple edges and lines

Layer 2

Corners, curves and textures

Layer 3+

Eyes, noses, ears and full faces

The deeper the layer, the more complex **the feature it learns to recognise.**

Real-world applications

CNNs power many technologies you use every single day

Face recognition

Unlocking your phone
and tagging photos

Medical diagnosis

Detecting cancer in
X-rays and MRI scans

Self-driving cars

Reading road signs
detecting pedestrians

Security cameras

Detecting intrusions
tracking movement

Image generation

GANs, style transfer
and deepfakes

Agriculture

Crop disease detection
via satellite imagery

When you use **Google Lens to identify** a plant, read a restaurant menu, or search by photo, a **CNN is doing all the work.**

You interact with **CNNs dozens of times a day.**

You just never knew what they were called.

CNN vs Regular Neural Network

Why not just use a regular neural network for images?

Here is why CNNs win

Regular neural network

Flattens image into one long list.
Loses all spatial information.

Every pixel connects to every neuron.
Millions of parameters needed.

Confused when same object
shifts position in the image.

Very slow and expensive
to train on image data.

CNN

Preserves spatial structure.
Knows where features appear.

Shared filters scan the whole image.
Far fewer parameters needed.

Recognises objects regardless
of where they sit in the image.

Efficient and designed specifically
for visual data processing.